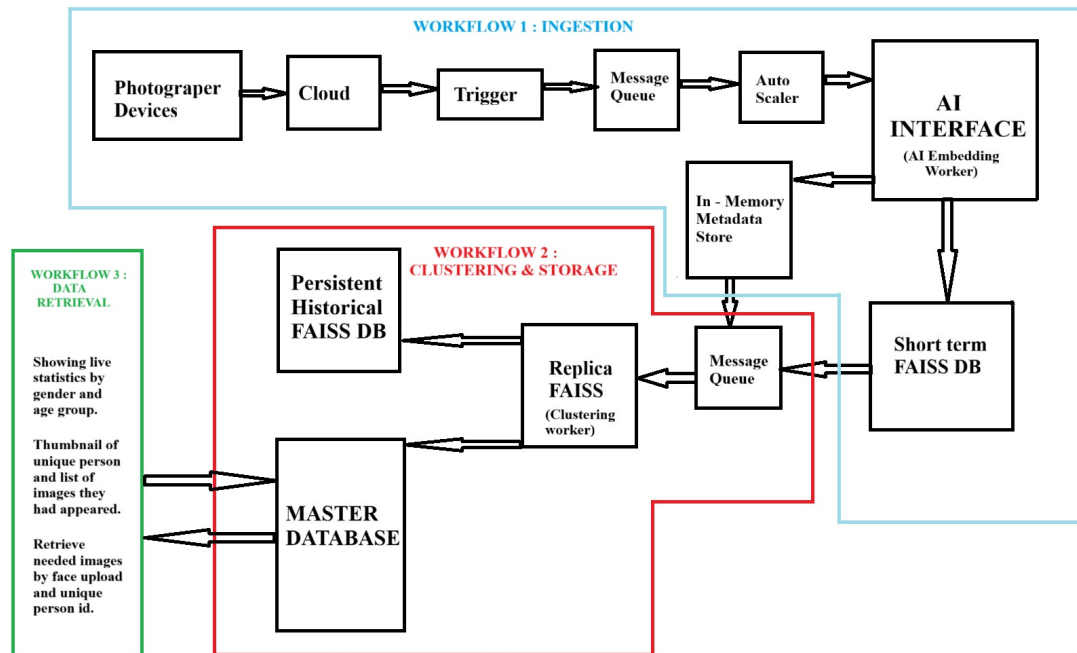


System Design Report

AI-Based Face Extraction and Clustering Pipeline

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AI-Based Face Extraction and Clustering Pipeline



1. Overview

This document provides a comprehensive architectural, operational, and technical overview of an AI-driven face extraction, clustering, and classification system built for live wedding photography. The system ingests thousands of photos in near-real time, detects and identifies unique individuals, classifies them by gender and age, and provides live visual analytics for event organizers or photographers.

The architecture is distributed, asynchronous, and fault-tolerant, with three independent workflows that communicate only via message queues and acknowledgments, ensuring decoupled scalability and resilience.

2. System Objectives

- Real-time ingestion of massive image uploads from multiple photographer devices.
- Automated AI-based face detection and embedding extraction using lightweight CNN or transformer-based facial encoders.
- Efficient clustering and deduplication of embeddings to identify unique individuals across photos and sessions.
- Persistent historical storage of embeddings, metadata, and classification details for future retrieval.
- Real-time dashboard and analytics interface for viewing statistics, thumbnails, and live presence summaries.
- Scalability and fault tolerance via message-driven architecture and auto-scaling compute workers.
- Workflow isolation ensuring that failures in one layer do not affect the others.

3. High-Level Architecture

The system comprises three independent workflows operating asynchronously:

1. Workflow 1 - Ingestion: Handles upload events, face detection, and embedding extraction.
2. Workflow 2 - Clustering & Storage: Clusters embeddings and persists unique identities.
3. Workflow 3 - Data Retrieval: Provides live analytics and dashboard visualization.

Each workflow runs in an isolated compute environment ensuring resilience and scalability.

4. Workflow 1 - Ingestion

Workflow 1 manages image intake, trigger functions, and AI workers that extract embeddings and classifications.

- Photographer uploads trigger batch events.
- Trigger functions group images into batches (25-75 images).
- Messages are sent to queues ensuring load decoupling.
- AI workers extract embeddings and classify attributes (gender, age).
- Embeddings stored in short-term FAISS and metadata stored in-memory.
- Upon batch completion, Workflow 1 sends BATCH_COMPLETED event to Workflow 2.
- After acknowledgment (BATCH_CLEARED), temporary data is purged for statelessness.

5. Workflow 2 - Clustering & Storage

Workflow 2 clusters embeddings, deduplicates identities, and persists unique persons.

- Consumes BATCH_COMPLETED from Workflow 1.
- Loads embeddings into Replica FAISS and metadata into local dictionary.
- Performs clustering (e.g., DBSCAN, Agglomerative) by gender/age filters.
- Updates Persistent Historical FAISS DB and Master Database.
- Sends acknowledgment (BATCH_CLEARED) back to Workflow 1.
- Clears temporary memory post-persistence to maintain efficiency.

6. Workflow 3 - Data Retrieval

Workflow 3 is read-only, providing visualization and querying features.

- REST/GraphQL endpoints serve dashboards and analytics.
- Caching layer accelerates query response.
- Master Database (read-only replica) used for queries and live counts.

Example queries:

```
SELECT gender, age_group, COUNT(*) FROM master_faces GROUP BY gender, age_group;  
SELECT * FROM master_faces WHERE person_id='P00042';
```

7. Data Flow Summary

Photographers -> Cloud -> Trigger -> Queue (WF1)

- > AI Workers -> Short-Term FAISS + Metadata
- > Message Queue -> Replica FAISS (WF2)
- > Cluster -> Persistent FAISS + Master DB
- > ACK -> Clear temporary data
- > Workflow 3 reads Master DB for dashboard display.

8. Reliability and Scaling

- Message Queues ensure asynchronous decoupling.
- Auto-Scaling adjusts to unpredictable workloads.
- In-memory metadata enables fast temporary processing.
- Clean acknowledgment mechanism maintains workflow integrity.
- Each workflow operates independently ensuring isolation.
- Centralized monitoring (ELK/CloudWatch) provides system observability.

9. Conclusion

The architecture achieves scalability, reliability, and operational independence.

Workflow 1 ensures real-time ingestion and embedding generation.

Workflow 2 guarantees deduplication and persistence.

Workflow 3 provides a seamless analytical interface for end users.

Workflow 3 exclusively connects to the Master Database ensuring security and isolation of FAISS resources.